

Software Engineering Project May-2026

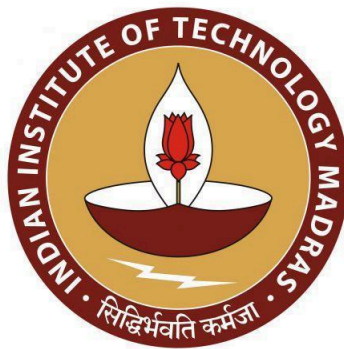
Milestone 5 Deliverables for Campus Connect

Final Submission

Submitted By

Team Name: NexMind

Team Code : Team-003



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Student Club & Event Management Platform for Higher-Education Campuses

Project Details

Field	Value
Software Name	Campus Connect
Problem Statement	Option 1: Community Services Platform
Subdomain	Student Club Management: Higher Education Institutions
Milestone	Milestone 5 – Final Submission
Scope	Complete working prototype, final compiled report (Milestones 1-5), tools and technologies used, issue-tracking record
Date of Submission	23-08-2026
Live Frontend	https://campus-connect-swe.vercel.app/
Live Backend	https://campusconnect.itshrestha.dev
Interactive API Documentation	https://campusconnect.itshrestha.dev/docs (Swagger UI)
Code Repository	https://github.com/Srivastava-Shrestha/MAY2026-Team-003

Team Details

Team Name: NexMind **Team Code:** Team-003

Name	Roll Number	Role in the Project
Atharv Khare	23f2004201	Lead Product Manager / Frontend Developer
Kavisha Tankle	23f1000041	Scrum Master / QA Engineer
Shrestha Srivastava	23f3000168	Backend Lead / Code Reviewer
Pawan Kumar Choudhary	22f3000162	Backend Developer / Tester
Shrishti Gupta	23f2004336	Frontend Developer

Team Responsibilities for Milestone 5

Member	Responsibilities in the closing milestone
Atharv Khare	Compilation of all final documentation and assets for submission; ownership of this report; integration of the AI Club Finder agent, acceptance testing and frontend bug fixes, final product sign-off
Kavisha Tankle	Coordination of the demo recording schedule; confirming the issue tracker reflects a clean, closed state before submission; final scrum record and project summary deck.
Shrestha Srivastava	Freezing the backend for submission; verifying that the deployed API, the OpenAPI specification and the submitted code zip agree; resolution of DOC-01, leaderboard and certificate APIs
Pawan Kumar Choudhary	Regression runs across the cumulative test suite; checking the backend and testing segments of the demo.
Shrishti Gupta	Freezing the frontend for submission; completion of the remaining module integrations; checking the student, leader and admin walkthrough segments

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1. Executive Summary

Campus Connect is a student club and event management platform built for higher-education institutions, developed by Team NexMind (Team-003) across five milestones from initial user research through to this final submission. The product replaces the fragmented mix of Instagram DMs, WhatsApp groups, Google Forms, spreadsheets and Canva that club leaders currently rely on with a single connected workflow: club discovery and membership, event publishing and registration, attendance and results, automatically generated and publicly verifiable certificates, club announcements, member issue tracking, an AI-powered club finder, and a campus-wide activity leaderboard.

The system is a Vue 3 single-page frontend against a FastAPI, async SQLAlchemy and PostgreSQL backend, deployed independently and integrated end to end. This report compiles the work across all five milestones into a single final document, alongside the tools and technologies used and the issue-tracking record maintained throughout the project.

Everything reported as delivered in this milestone was verified against live infrastructure rather than assumed working: password recovery was confirmed with a real email through production SMTP, and the AI Club Finder and certificate module were exercised against real seeded campus data rather than a mock dataset.

2. User Identification and User Stories

User identification and the initial user-story backlog were established in Milestone 1 through interviews with a student, two club leaders, and a written submission from a student body representative. The personas and stories below are carried forward unchanged in substance from that research and reflect what was actually built by the end of Milestone 5.

2.1 Primary Users

Students and club members (for example Shikha Singh, undergraduate student, KNIT Sultanpur). The widest user group on the platform: they discover clubs, send join requests, register for events, check in, receive results, download verifiable certificates, and raise issues about their own participation. Every workflow on the platform is ultimately designed around this experience.

2.2 Secondary Users

Club heads and event coordinators (for example Aayansh Yadav, former head of the campus gaming club; Aayush, campus ambassador, independent builder club). They create and run clubs and events, review join requests, mark attendance, declare results,

post announcements, and resolve member issues – the administrative load the platform is built to remove.

Campus council administrators and student affairs coordinators. The institutional gatekeeper: reviews and approves or rejects new club registrations, and configures the college's verified student email domain on the platform.

2.3 Tertiary Users

Student body representatives and institutional administration (registrar, principal, dean of student affairs). They do not use the platform directly, but their feedback on credibility, verifiable records and student accountability shaped the recognition features – certificates and the club activity leaderboard – built into the product.

2.4 User Story Epics

The full backlog of eighteen user stories across seven epics is documented in the Milestone 1 deliverable and is not reproduced in full here. The epic structure below summarises what each covers and its delivery status as of this final submission.

Epic	Covers	Status at Milestone 5
1. Platform Foundation	College registration, signup and login against a verified college email domain, password recovery and Google Sign-In	Delivered
2. Club Discovery and Membership	Club directory, club profiles, join requests, membership approval, club registration and campus-admin approval	Delivered
3. Event Lifecycle	Event creation and publishing, registration, attendance, results (winner / runner-up / participant), edit and cancel	Delivered
4. Communication and Issue Tracking	Club announcements, a consolidated announcement feed, member-raised issues and leader resolution	Delivered
5. Certificate Workflow	Automatic PDF certificate generation on result declaration, a student certificate wallet, public serial-based verification	Delivered
6. AI-Powered Club Discovery	Natural-language club matching against real college data via a bounded tool-calling agent	Delivered
7. Recognition and Engagement	Campus-wide club activity leaderboard ranked by events run, membership growth and participation	Delivered

3. Milestone Compilation

This section compiles the work delivered across all five milestones into one continuous record, from initial research through to this final submission.

3.1 Milestone 1 – Problem Definition and User Research

Established the problem – fragmented, manual club operations spread across informal tools – and identified primary, secondary and tertiary users through three interviews and one written submission. Produced the initial backlog of eighteen user stories across seven epics that the rest of the project was built against. A future-scope list (AI-assisted drafting, cross-college federation, budget tracking and others) was recorded so that out-of-scope ideas could not re-enter the sprint backlog later.

3.2 Milestone 2 – Design and Foundation

Delivered the system design: the entity-relationship model, the API surface, and the visual design system – a warm, role-accented interface (orange for students, green for club leaders, navy for administrators) intended to feel approachable to students while remaining credible to institutional staff. Laid the technical foundation both sub-teams built on for the remainder of the project: the FastAPI and PostgreSQL backend structure, and the Vue 3 and Pinia frontend shell.

3.3 Milestone 3 – Sprint 1: Core Participation Spine

First development sprint. Delivered the participation spine of the product: club registration and admin approval, membership join requests, event publishing and registration, attendance marking and per-registration results. Testing surfaced four functional defects (BUG-01 to BUG-04), all resolved before Milestone 4, and one documentation gap – DOC-01, the generated OpenAPI specification not declaring its error responses – which was carried forward as a tracked issue rather than quietly closed.

3.4 Milestone 4 – Sprint 2: Certificates, Auth Recovery and Integration

Second development sprint, directly responsive to feedback gathered at the Sprint 1 demo from the same users interviewed in Milestone 1. Delivered the certificate module (automatic PDF generation, student certificate wallet, public verification by serial number), replaced the per-registration result flow with a single default-and-override declaration endpoint (F-02), added an event reminder notification (F-03), and delivered password recovery end to end (F-06). The cumulative backend test suite reached 459 passing cases. Three items were explicitly carried into Milestone 5: bulk check-in with a searchable roster (F-01), an announcement seen count (F-04), and the DOC-01 specification fix (F-05).

3.5 Milestone 5 – Final Submission

Closed out the carried-over Milestone 4 items and completed the remaining epics against the live backend rather than mock data. Delivered in this milestone:

- Google Sign-In – the second half of the recovery and access request made by the primary student user at the Sprint 1 demo, alongside the completed forgot and reset password flow verified with a real email through production SMTP.
- The club activity leaderboard (Story 7.1), the tertiary users' highest-value ask and the item campus administration repeated at both demos.
- The AI Club Finder (Story 6.1), rebuilt as a bounded tool-calling agent operating against real club and event data rather than a mock corpus.
- Real file uploads for club banners and profile pictures to S3 storage.
- Frontend integration of the modules that were backend-complete but unwired at Milestone 4 – notifications, attendance, results and certificates – together with an application-wide UI and empty-state consistency pass across the student, club-leader and admin dashboards.

This closes the gap the team reported openly in Milestone 4 §11, where four coupled modules were held back from integration rather than wired hurriedly at the end of the sprint.

4. Tools and Technologies Used

Layer	Tools / Technologies
Frontend	Vue 3 (Composition API), Vite, Pinia, Vue Router, vanilla CSS design-token system
Backend	FastAPI, Python 3.12+, async SQLAlchemy, Alembic migrations, Pydantic
Database	PostgreSQL (Neon serverless Postgres with connection pooling)
Authentication	JWT (access, refresh and single-use reset tokens), Google Identity Services (OAuth)
AI / LLM	Anthropic Claude (Haiku 4.5) - bounded tool-calling agent for the AI Club Finder
File Storage	AWS S3 (club banners, profile pictures, certificate PDFs)
Email	SMTP via Resend for password-reset delivery; Jinja2 HTML email templates
Certificate Generation	Server-side PDF rendering from SVG templates per result type, with an embedded verification QR code
Testing	pytest, pytest-asyncio, a disposable per-run PostgreSQL schema for integration tests
Package Management	uv (backend), npm (frontend)
CI / Deployment	GitHub Actions (backend test workflow), Docker, Vercel (frontend), Render (backend)
Issue Tracking	GitHub Issues - see §5
Project Tracking	Trello (Kanban board), three-tier Git branch strategy (main → dev → feature/name-task)
Documentation	OpenAPI 3.1.0 and Swagger UI, generated directly from the FastAPI routes and Pydantic models

5. Issue Tracking

The team tracked defects, feedback items and carried-over work as GitHub Issues on the project repository throughout development rather than in an external tracker. Each issue was labelled by its milestone of origin and closed with a reference to the commit or pull request that resolved it, which is how traceability between a user-feedback item (for example F-01 to F-06 in Milestone 4) and its resolution was maintained across sprints.

Milestone	Issues logged	Closed by end of milestone	Carried forward
Milestone 3 (Sprint 1)	4 functional defects (BUG-01 to BUG-04) and 1 documentation defect (DOC-01)	4	1 (DOC-01)
Milestone 4 (Sprint 2)	Feedback items F-01 to F-06 plus regressions and integration issues (#50, #54, #56)	7	3 open at submission, plus F-01, F-04 and F-05 / DOC-01

Milestone	Issues logged	Closed by end of milestone	Carried forward
Milestone 5 (Final)	Carried items closed out; final polish and bug reports tracked as issues during the closing pass	All carried items closed	0

The screenshot below shows the issue board as it stood during the closing pass, with the three items carried out of Milestone 4 – the red frontend test suite (#56), the member profile endpoint (#54) and the remaining UI-blocking endpoints (#50) – all of which were closed before this submission.

Note that related issues were added in sub-issues so total closed issues on github are 9

All issues New issue

is:issue state:closed

☐ Open **0** **Closed 9** Author Labels Projects Milestones Assignees Newest

- ☐ ☒ **CORS blocks local frontend from testing against deployed API** help wanted
#55 · by 1mystic was closed 2w ago 1 comment
- ☐ ☒ **Endpoint to view another member's profile** backend bug enhancement
#54 · by 1mystic was closed 3d ago 1 comment
- ☐ ☒ **Auth: verify email, forgot password and reset password endpoints missing** backend bug enhancement
#53 · by 1mystic was closed 2w ago 1 comment
- ☐ ☒ **POST /auth/onboarding is missing : student onboarding data is currently discarded** bug
#52 · by 1mystic was closed 2w ago 1 comment
- ☐ ☒ **Public endpoint for landing page club carousel** bug
#51 · by 1mystic was closed 2w ago 1 comment
- ☐ ☒ **Backend endpoints required to complete UI (follow-up to #46)** 5/5 backend
#50 · by 1mystic was closed 3d ago 1 comment
- ☐ ☒ **UI issues after backend integration**
#46 · by Srivastava-Shrestha was closed 2w ago 1 comment
- ☐ ☒ **Auth Testing**
#19 · by Srivastava-Shrestha was closed 1mo ago 2 comments
- ☐ ☒ **Move pages Directory into Frontend and Use a Separate Feature Branch**
#4 · by Srivastava-Shrestha was closed on Jul 2 1 comment

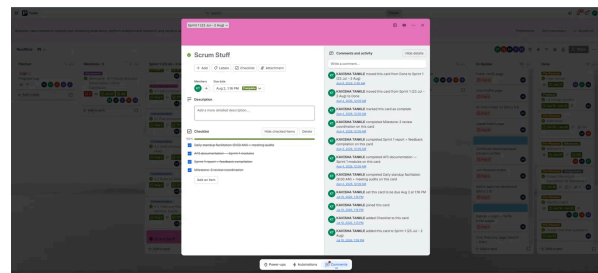
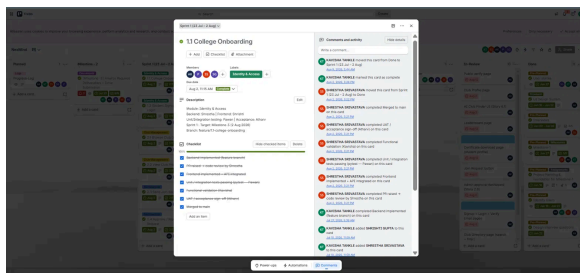
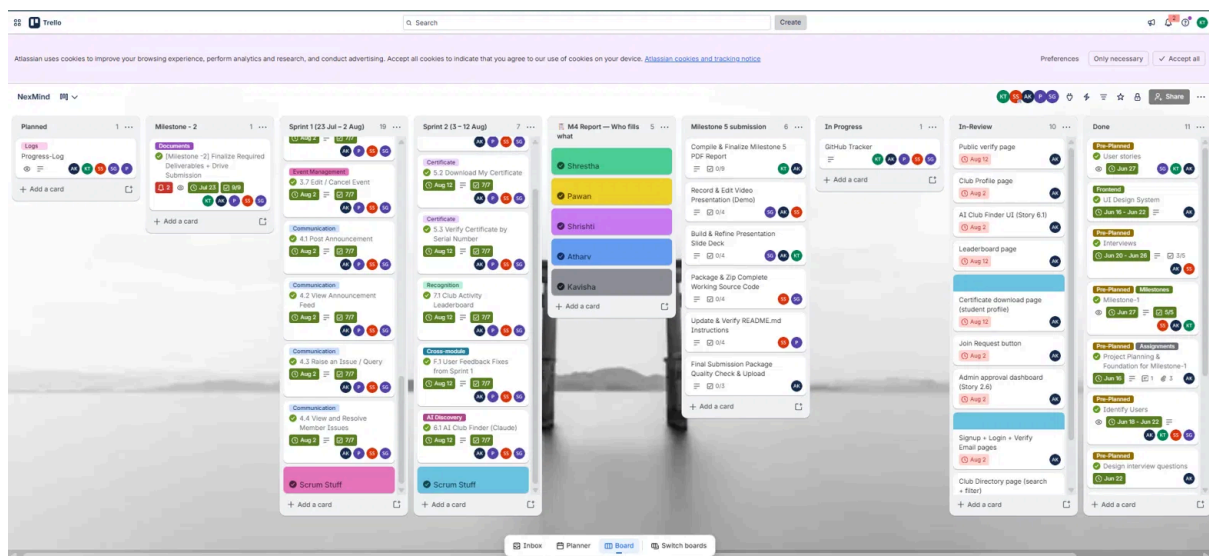
Figure 1: GitHub Issues board during the Milestone 5 closing pass

FINAL TRELLO BOARD :

Link - <https://trello.com/b/M4qafINT/nexmind>

Sprint planning and tracking were managed through Trello rather than Jira. The user story map was converted directly into Trello story cards, one card per user story, with a module label (Identity, Events, Communication, Membership, Certificates, etc.), an assigned owner, a due date matching the sprint deadline, a checklist, and a linked Git branch name in the format `feature/<story-id>-<short-name>`, so each card mapped directly to its corresponding feature branch.

Sprint 1 (23 Jul – 2 Aug) was tracked with 19 story cards; Sprint 2 (3–12 Aug) had 5, covering the certificate module and fixes carried over from Sprint 1 client feedback. Scrum Master tasks – standups, functional validation across all cards, documentation, and sprint reporting – were tracked separately through process cards rather than story cards, since that work wasn't tied to a single user story.



6. System Architecture

The four diagrams below describe the system as it stands at final submission: how the pieces are deployed and talk to each other, what the data model looks like, how the central participation workflow runs end to end, and how the AI Club Finder agent is bounded.

6.1 High-Level System Architecture

The browser runs a Vue 3 single-page application that talks to the FastAPI backend over HTTPS with a JWT bearer token. Inside the backend, REST routes delegate to a service layer, which reaches the database through a repository layer, so persistence concerns never leak into the route handlers. Four external services sit outside that boundary: Google Identity Services for OAuth sign-in, Resend for transactional email, Anthropic Claude for the agent's model calls, and AWS S3 for banners, avatars and certificate PDFs. The Anthropic API key is held server-side only - the browser never sees it.

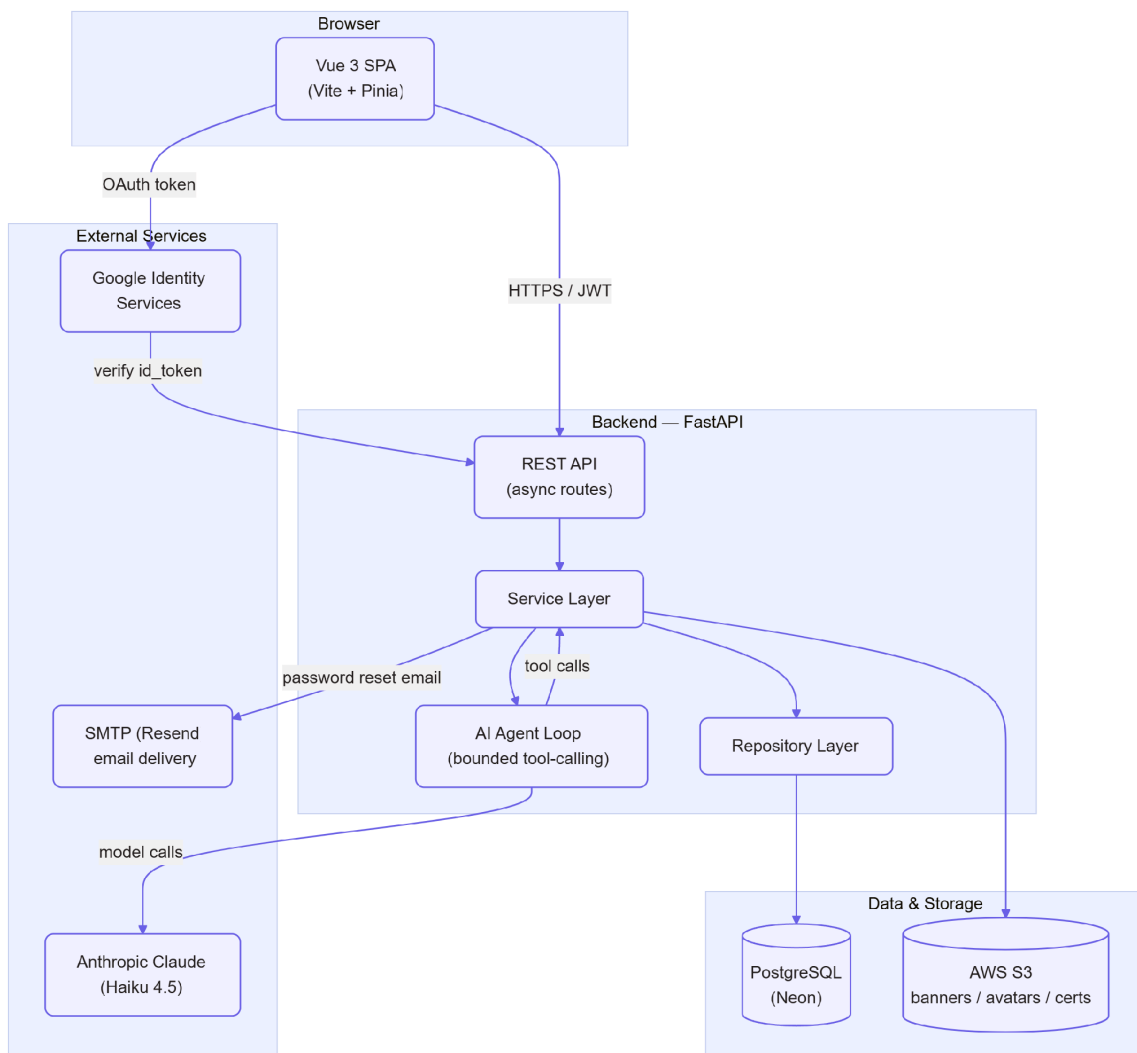


Figure 2: High-level system architecture – browser, FastAPI backend, external services and storage

6.2 Entity-Relationship Diagram

Every entity is scoped to a college, which is what keeps one institution's data invisible to another. A user is scoped to a college and specialises into a student; a student holds memberships, raises issues, receives announcements and registers for events. Clubs belong to a college and own their announcements, links and events. The participation spine runs down the right-hand side: an event has registrations, a registration carries the check-in flag and the recorded result, and a certificate hangs off the registration that earned it, which is why a certificate can always be traced back to a specific person at a specific event.

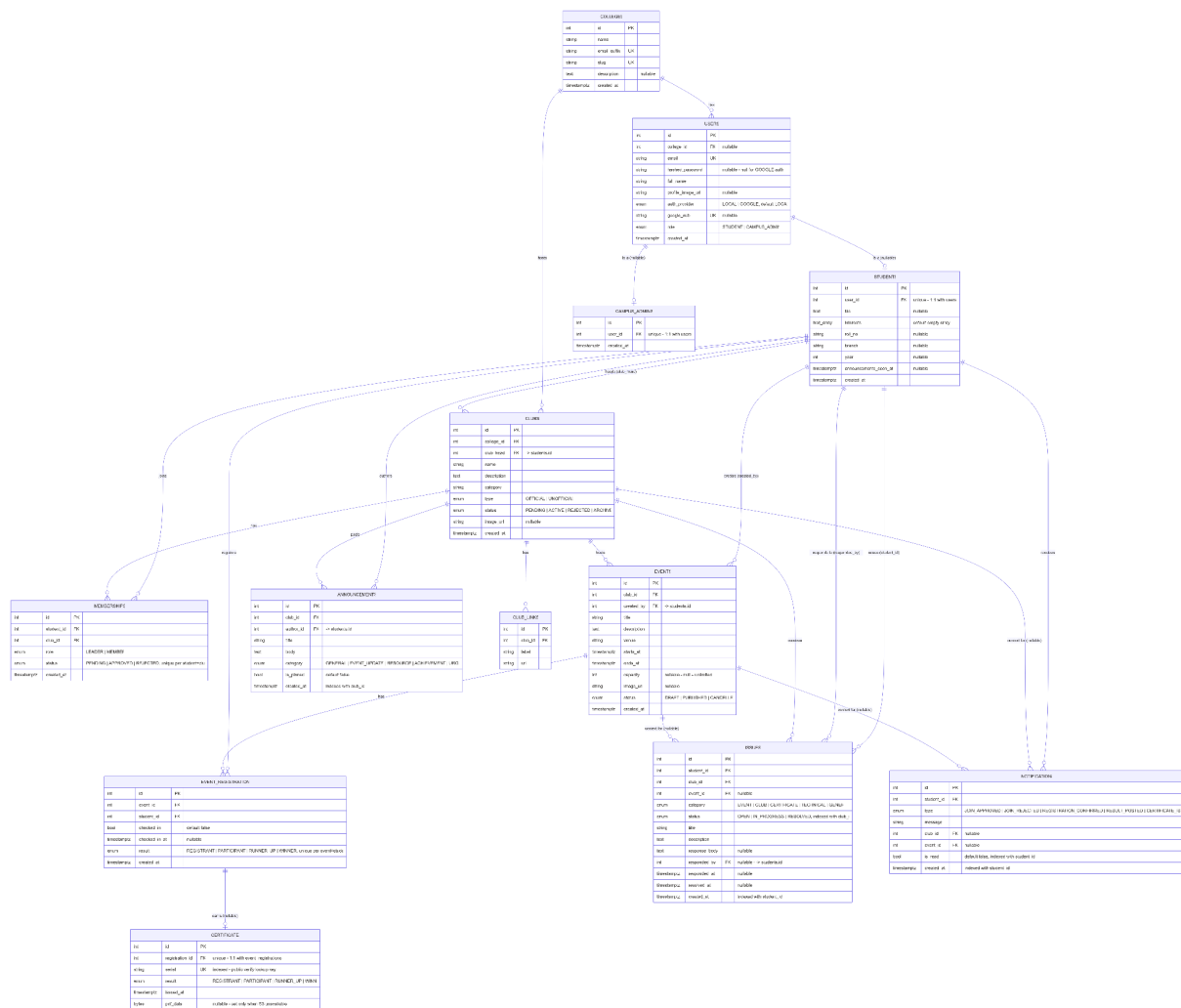


Figure 3: Campus Connect entity-relationship diagram

Entity / Area	Old ERD	New ERD	Change
COLLEGES	Basic college data	Same	Mostly unchanged
USERS	Basic auth/user fields	Google auth support	Expanded
STUDENTS	Basic student profile	Interests + announcement tracking	Expanded
CAMPUS_ADMINS	Basic 1:1 user profile	Same + explicit 1:1	Minor
CLUBS	created_by, approved_by	club_head	Model changed
CLUB_LINKS	Same	Same	Unchanged
MEMBERSHIPS	Membership	Explicit unique student/club	Constraint added
EVENTS	Has embedding	Has status, no embedding	Changed
EVENT_REGISTRATIONS	Registration	More constraints/defaults	Expanded
CERTIFICATES	Certificate metadata	Adds PDF storage	Expanded
ANNOUNCEMENTS	Body only	Title/category/pinning	Expanded
ISSUES	Absent	Full issue/ticket system	NEW ENTITY
NOTIFICATIONS	Student notifications	Club/event contextual notifications	Expanded
Timestamps	datetime	timestamptz	Standardized
Indexes	Mostly not shown	Multiple indexes documented	Added
Unique constraints	Mostly not shown	Membership + registration uniqueness	Added

6.3 Core Request Sequence – Event Registration to Certificate

This is the participation spine as a single runtime sequence. A student registers and the backend creates an EventRegistration. On event day the leader marks attendance, flipping checked_in, and then declares results in one call. That single call sets the winner, the runner-up and every remaining attendee as a participant, renders a certificate PDF per attendee, uploads each to S3, records the certificate with its serial and URL, and notifies the student. Nothing between the result declaration and the certificate requires a human step.

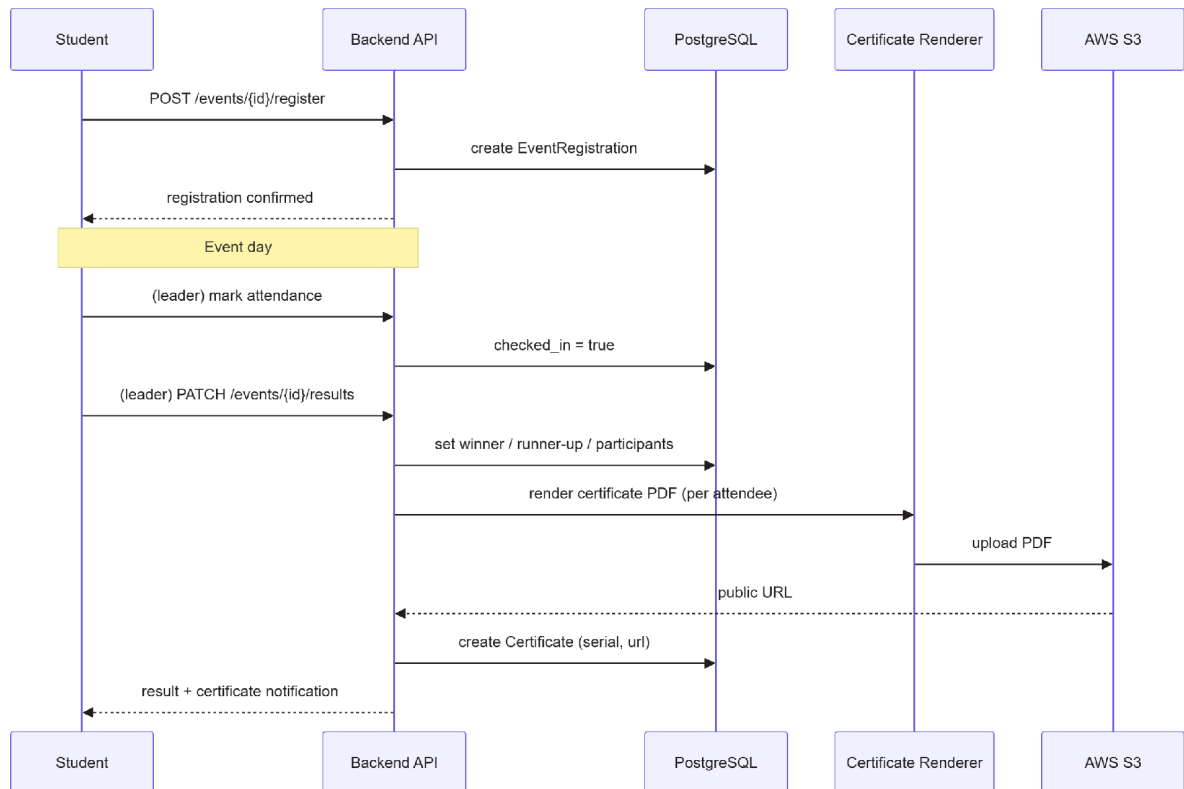


Figure 4: Sequence – register, attend, declare results, certificate issued

6.4 AI Club Finder – Agent Loop

The AI Club Finder is the product's only GenAI feature, and it is deliberately bounded. The model may either answer or call a tool; tool results are appended to the conversation and the loop runs again. Three limits close the loop: a budget on iterations, tool calls and tokens that forces a final answer from whatever has been gathered; a grounding gate that permits the reply to name only entities a tool actually returned, so the agent cannot invent a club; and a deterministic recommender that serves as a degraded fallback if the model is unreachable. The student therefore always receives an answer, and that answer is always drawn from real college data.

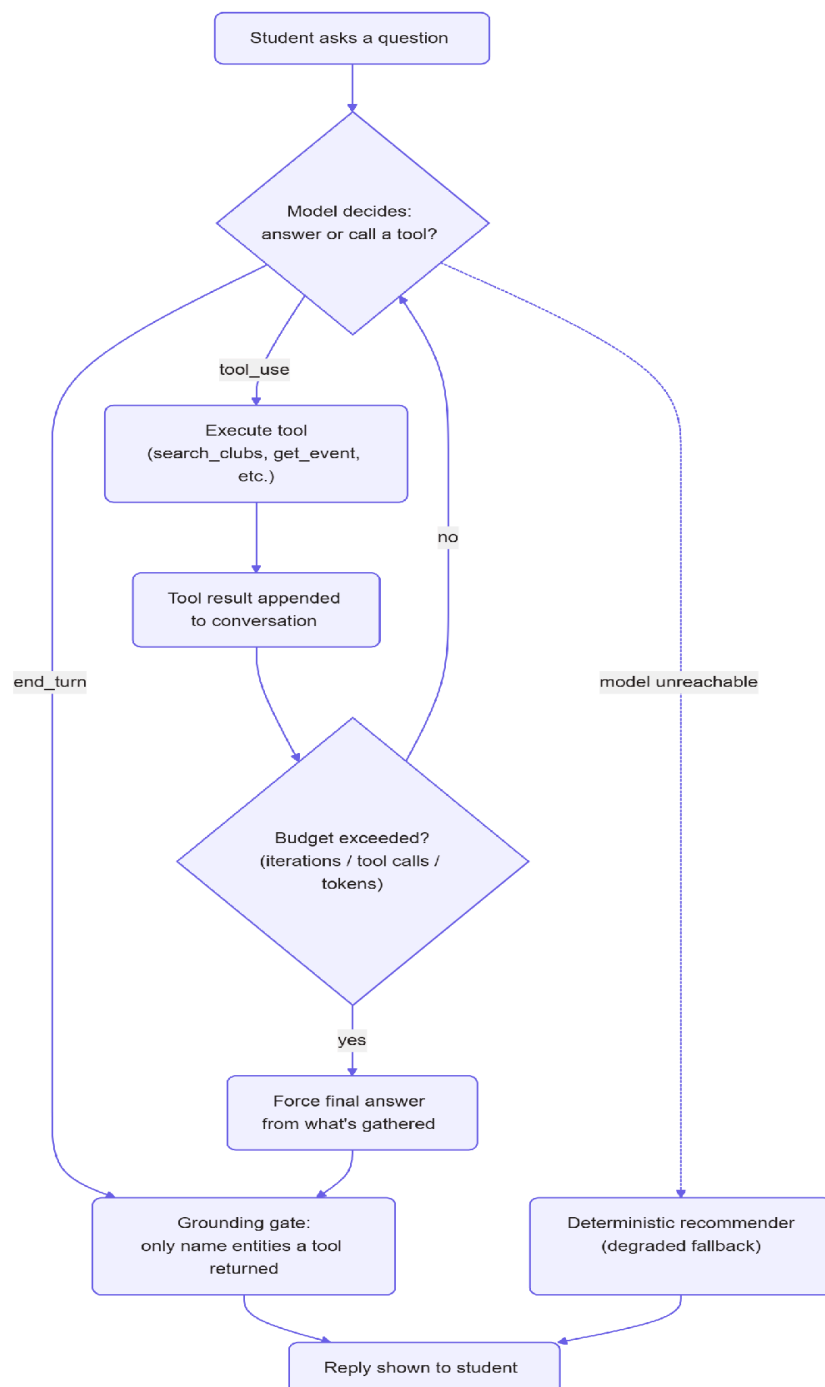


Figure 5: AI Club Finder - bounded tool-calling agent loop with grounding gate and deterministic fallback

7. Final Submission Artefacts

The checklist below reflects the deliverables required for Milestone 5 as specified by the course.

Deliverable	Status / Location
Video presentation - project overview, user stories and a working demo	https://drive.google.com/file/d/177m9lovE-NBMVcLbbEQL9zbQSkSnx0x7/view?usp=sharing
Complete project code (zip)	https://drive.google.com/file/d/18VDU0nii6LvAG4Ulfq5U9svOncmzR0hE/view?usp=sharing
README with setup and run instructions	Included at the root of the submitted code zip and in the repository
Final report - compilation of Milestones 1 to 5	This document
Tools and technologies used	§4 of this document
Issue tracking record	§5 of this document
System architecture and data model	§6 of this document
Live frontend	https://campus-connect-swe.vercel.app/
Live API and interactive documentation	https://campusconnect.itshrestha.dev/docs